

# FedAGC: Federated Continual Learning with Asymmetric Gradient Correction

Chengchao Zhang<sup>1</sup>, Fanhua Shang<sup>1†</sup>, Hongyin Liu<sup>2†</sup>, Liang Wan<sup>1,2</sup>, Wei Feng<sup>1</sup>,

<sup>1</sup>College of Intelligence and Computing, Tianjin University, China

<sup>2</sup>Medical College, Tianjin University, China

{chengchaozhang, fhshang, hylu2009, lwan, wfeng}@tju.edu.cn

## Abstract

Federated Continual Learning (FCL) has emerged as a prominent distributed learning paradigm and aims at addressing model learning challenges in both federated and continual learning settings. Efficient personalization in FCL remains a major challenge, as it must handle not only conflicts between old and new knowledge within parallel task streams but also heterogeneous knowledge conflicts from different clients. Recent approaches attempt to mitigate these issues through gradient correction. However, they often overlook the combined impact of gradient magnitude and direction, leading to unsatisfactory gradient solutions. To address these issues, we propose a novel federated continual learning method (called FedAGC) with asymmetric gradient correction, which performs memory rehearsal using representative samples selected via a centroid-based approach from historical tasks. By formulating the problem as a multi-objective optimization problem, FedAGC derives more effective gradients while incorporating group-level personalization to facilitate useful knowledge integration and irrelevant knowledge isolation, effectively mitigating both temporal and spatial catastrophic forgetting. Extensive experiments confirm the effectiveness of FedAGC.

## 1. Introduction

Federated Learning (FL) [21, 22, 29], as a novel distributed machine learning paradigm, aims to train a global model by leveraging independent and diverse data across clients in a privacy-preserving and collaborative manner. FL has been widely applied in various domains, such as medical image analysis [23] and object detection [24]. It has been proven to effectively address the issue of data silos [33] without compromising client privacy. Most existing FL methods are limited to single-task learning, assuming that client data remains static over time. However, in real-world scenarios, data is continuously evolving, forming sequential tasks with

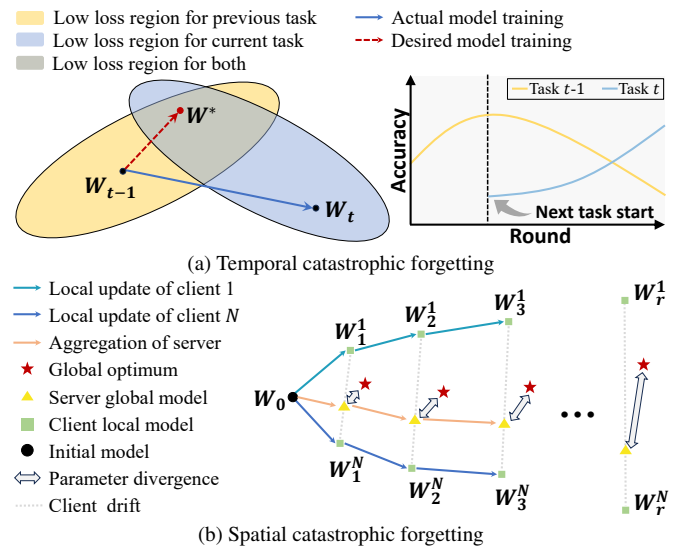


Figure 1. *Spatial-Temporal Catastrophic Forgetting*. (a) Direct access to new tasks leads to a significant performance decline on old tasks. (b) Non-IID data exacerbates the weight disparity.

unpredictable future changes. As shown in Fig. 1a, severe temporal catastrophic forgetting (TemporalCF) [38, 45] occurs when the local model on each client learns a newly introduced task, leading to performance degradation on previously learned tasks as new knowledge is acquired.

Improving the generalizability of FL in multi-task evolving scenarios has emerged as a critical and challenging research topic. Recently, Continual Learning (CL) [17] has made significant advances in addressing TemporalCF, enabling effective mitigation of this problem. Based on such characteristics, the concept of Federated Continual Learning (FCL) [6, 46, 47] came into being, which integrates CL methods into the FL framework. In the FCL setting, FL can be regarded as a distributed instance of conventional CL, while CL methods assist in addressing FL challenges.

FCL breaks the static limitation of traditional FL, where each client is confined to learning a single task sequence. However, integrating CL into the FL framework poses substantial challenges. Besides TemporalCF, FCL also

<sup>†</sup>Corresponding author.

faces spatial catastrophic forgetting (SpatialCF). Specifically, the private data of different clients in FCL may follow drastically different distributions. This situation, where data across clients is Non-Independent and Identically Distributed (Non-IID), which is referred to as statistical heterogeneity [26]. During server-side aggregation, where gradients from multiple clients are combined to initialize the next round of local CL training, statistical heterogeneity can lead to issues such as parameter divergence and client drift [14, 37], as illustrated in Fig. 1b. Consequently, the aggregated global model exhibits reduced generalizability, further exacerbating temporal catastrophic forgetting.

To address the aforementioned challenges, recent studies [42, 47, 48] have explored the problem of Personalized Federated Continual Learning (PFCL). Unlike traditional FCL that aim to obtain a universal global model, PFCL leverages efficient server-side aggregation to facilitate personalized training. Existing PFCL methods mainly focus on optimizing local models for clients, ensuring strong performance on both new and previous tasks in a parallel, independent, and heterogeneous task stream across clients. The most relevant work to ours is FedMes [42], which builds on the gradient correction technique of A-GEM [2] and enhances local models through KNN-based inference. However, existing methods typically rely solely on gradient direction to determine similarity, ignoring the impact of magnitude, leading to suboptimal gradients. In contrast, we propose a new Asymmetric Gradient Correction (AGC) technique, which adaptively balances the combined impact of direction and magnitude, enabling a smoother optimization transition.

We formulate the PFCL problem as a Multi-Objective Optimization (MOO) problem to obtain the optimal gradient, where each objective minimizes the distance to a target gradient. Specifically, to mitigate Spatial-Temporal Catastrophic Forgetting (STCF), each client extracts a small set of representative samples after completing local training for each task and employs them for rehearsal in subsequent learning. Subsequently, we correct the gradients by considering both their magnitudes and directions, ultimately obtaining an optimal gradient that does not damage any of old or new knowledge at each update from a set of class-level gradients. Additionally, empirical evidence suggests that a single global model has limited generalization across all clients [34]. Therefore, we introduce a group-level personalization strategy, where clustering is performed based on the similarity of task gradients uploaded by participating clients. These clusters are then divided into groups, enabling joint training of multiple global models.

Our main contributions can be summarized as follows:

- To the best of our knowledge, this is the first attempt to unify both spatial and temporal catastrophic forgetting in PFCL through a multi-objective optimization framework.
- We propose a novel asymmetric gradient correction algo-

rithm that performs memory rehearsal using representative samples selected via a centroid-based approach from historical tasks, transforming multi-task updates into an optimal gradient search problem that minimizes inter-task discrepancies and mitigates TemporalCF.

- We propose a new group-level personalization approach that clusters clients by gradient similarity and aggregates multiple global models, isolating irrelevant knowledge interference while fostering the positive fusion of related knowledge to mitigate SpatialCF.
- Our method outperforms all the baselines in terms of accuracy and forgetting rate, demonstrating its efficacy in addressing the PFCL problem.

## 2. Background and Related Works

**FL Algorithms for Statistical Heterogeneity.** The client-server architecture-based FL paradigm is currently the most widely adopted framework. This distributed learning model involves a central server for aggregating network model parameters or gradients and  $N$  clients  $\{F_1, F_2, \dots, F_N\}$  for participating in the training of local private data, where each client possesses a local private dataset  $\{D_1, D_2, \dots, D_N\}$ . For the  $i$ -th client  $F_i$ , its local dataset includes  $n_i$  samples, denoted as  $\{D_i = \{(X_k^i, Y_k^i)\}_{k=1}^{n_i}\}$ . Different clients may exhibit distinct features and labels, such that  $X_i \cap X_j = \emptyset$  for  $\forall i, j \in [1, N]$ , and  $Y_k \neq \emptyset, Y_i = \emptyset$  for  $\forall i \in [1, N] \setminus \{k\}$ . The objective of FL is to collaboratively and privately train a global statistical model across multiple distributed clients under the supervision of a central server. The key formulation controlling FL in their classical FedAvg [29] is as follows:  $\min_{w \in \mathbb{R}^d} f(w) = \frac{1}{m} \sum_{i=1}^m f_i(w)$ , where  $f_i(w) = \mathbb{E}_{(x_i, y_i) \sim \mathcal{P}_i} [\mathcal{L}_i(w, x_i, y_i)]$  represents the local loss function of the  $i$ -th client based on its local data, and  $\mathcal{P}_i$  denotes the local data distribution of the  $i$ -th client. To address the challenge of statistical heterogeneity and mitigate client drift caused by Non-IID data distributions, FedProx [19] extends FedAvg by introducing a proximal term that effectively constrains local updates, thereby enhancing the overall stability of model convergence. Another representative approach, SCAFFOLD [14], employs control variates to alleviate client drift. However, under extreme Non-IID conditions, these methods often struggle to produce a highly generalizable global model. Consequently, Personalized FL [20, 45] shifts its focus to providing specialized local models for each client, enabling personalized services. Per-FedAvg [7] combines meta-learning techniques during a pre-training phase to obtain a well-initialized model. While these approaches have demonstrated strong performance in FL, they are primarily designed to address SpatialCF and may not be sufficient for more complex scenarios.

**Gradient Correction-Based CL Algorithms.** In contrast to traditional FL designed for single-task scenarios, in a standard continual learning setting, there are a sequence

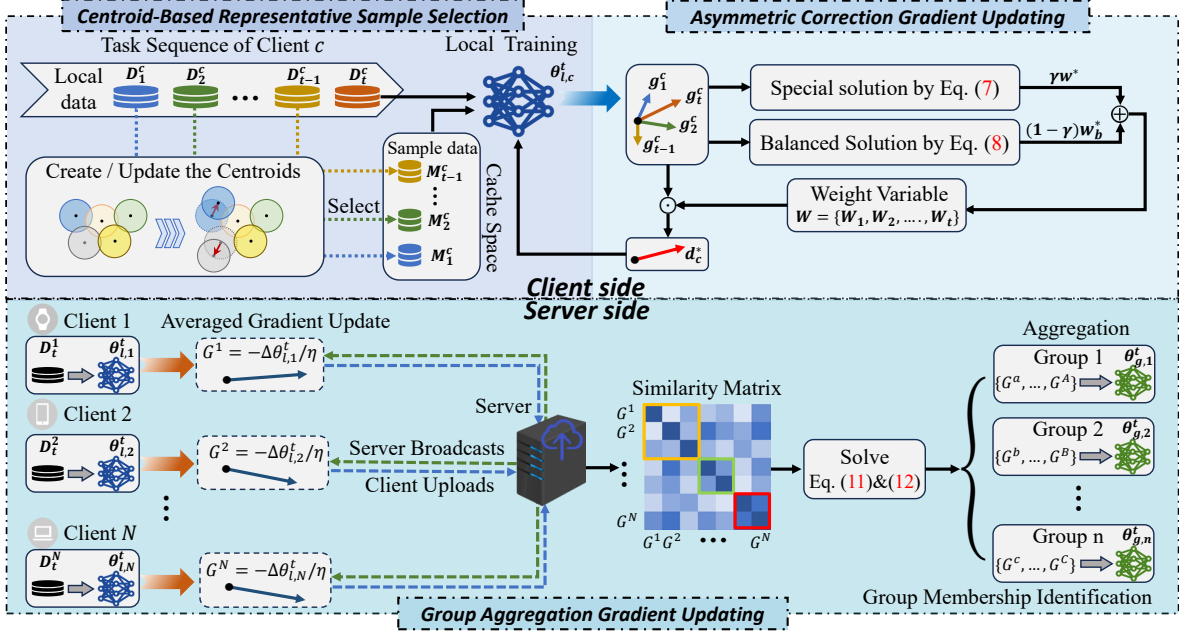


Figure 2. The workflow of the proposed FedAGC method. Our FedAGC consists of the following three modules: On the client side, the *Centroid-based Representative Sample Selection* (CBRSS) module rehearses the gradients of previous tasks to assist the *Asymmetric Correction Gradient Updating* (ACGU) module in generating optimal gradients. On the server side, the *Group Aggregation Gradient Updating* (GAGU) module clusters clients based on gradient similarity and aggregates multiple global models.

of tasks  $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T\}$ , where  $T$  is the number of tasks. For the  $t$ -th task  $\mathcal{T}_t$ , its dataset is composed of  $n_t$  pairs of data and labels:  $\{x_i^t, y_i^t\}_{i=1}^{n_t}$ . Typically,  $\mathcal{T}$  refers to a task stream where the arrival order of new tasks is unknown. As a result, the model can only access  $\mathcal{T}_t$  during the training of the  $t$ -th task, and access to it is no longer available afterward. Therefore, for  $\mathcal{T}_t$ , given the set of model parameters  $w^{(t)}$  learned so far, the learning objective at task  $t$  is:  $\min_{w^{(t)}} \mathcal{L}(w^{(t)}; w^{(t-1)}, \mathcal{T}^t)$ . In recent years, numerous CL algorithms have adopted gradient correction strategies to mitigate TemporalCF, typically involving two key steps: *Memory Rehearsal* and *Optimal Gradient Computation* [15, 27]. A representative example is GEM [25], which retrieves gradients from previously stored memories and integrates them with the current task's gradient to obtain an update direction suitable for all tasks. As one variant, A-GEM [2], consolidates multiple memories to enhance computational efficiency. Meanwhile, ER [3] combines data from episodic memory with the current task during training. Although ER is simple and effective, its performance depends heavily on the quality of the stored memories. Another related approach is parameter regularization [5, 8], which introduces additional regularization terms to consolidate knowledge from previous tasks and reduce interference between new and old task parameters. Among these methods, EWC [38] employs the Fisher information matrix to quantify the importance of parameters for previously learned tasks.

While FL and CL have achieved remarkable success in

their respective domains, effectively mitigating the STCF problem in PFCL remains a challenging endeavor [12, 45]. FedCurv [38] incorporates EWC to handle statistical heterogeneity; however, it incurs high computational overhead and lacks a dedicated mechanism for addressing TemporalCF. FedKNOW [44] deploys a gradient integrator at each client to reduce negative knowledge transfer by effectively combining signature tasks identified from past and current tasks of other clients through the global model. Recent advances in vision foundation models have driven an alternative prompt-based approach [9, 11, 43, 48] that mitigates STCF by adapting to new tasks through training only a small subset of prompts and classifier parameters.

### 3. The Proposed Method

In this section, we elaborate on the proposed FedAGC. For the first time, FedAGC formulates the PFCL problem (see Sec. 3.1) as a MOO problem. On the client-side, after training each task, we select representative samples to construct a memory cache (see Sec. 3.2) and use them to perform asymmetric gradient correction for the subsequent new tasks (see Sec. 3.3), thus avoiding TemporalCF. On the server-side, we group the average gradient updates uploaded by the clients (see Sec. 3.4) and aggregate multiple global models to avoid SpatialCF, while also satisfying the personalized local requirements.

The overall framework of FedAGC is shown in Figure 2, the algorithm is summarized in Algorithm 1.

### 3.1. Personalized Federated Continual Learning

We extend the traditional CL and FL methods to PFCL, which enables the independent and parallel integration of multiple task sequences across clients. This approach achieving both the fusion of local knowledge from new and old tasks and the integration of heterogeneous task knowledge across clients. Unlike FCL, which aims to obtain a general model that integrates global knowledge, PFCL addresses the diverse local needs of clients in highly heterogeneous scenarios, resulting in personalized local models. Specifically, each client learns its local model by optimizing the following objective:

$$\min_{w^{(t)}} \mathcal{L}(w^{(t)}; \mathcal{T}^t) + \frac{\lambda}{2} \|w^{(t)} - w_0\|^2, \quad (1)$$

where  $\mathcal{L}(\cdot)$  represents the empirical risk loss,  $\lambda$  is the regularization coefficient, and  $w_0$  denotes the initial model.

The PFCL scenario we consider, involving independent parallel multi-task sequences, comprises three key steps:

(1) Given  $N$  clients  $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_N\}$  and a server  $\mathcal{S}$ . There are task streams  $\{\mathcal{T}^1, \mathcal{T}^2, \dots, \mathcal{T}^N\}$ , within which each client  $\mathcal{C}_c$  engages in training a local CL model on its unique, private, and accessible task sequence  $\mathcal{T}^c = \{\mathcal{T}_1^c, \mathcal{T}_2^c, \dots, \mathcal{T}_T^c\}$ .  $\mathcal{T}^c$  consists of  $T$  distinct tasks, where the  $t$ -th task  $\mathcal{T}_t^c$  includes  $K_t^c$  classes, and the total set of classes for  $\mathcal{C}_c$  is represented as  $K^c = \{K_1^c, K_2^c, \dots, K_T^c\}$ .

(2) For the current task  $\mathcal{T}_t^c$  being learned, the participating client receives the model  $\theta_g^t$  globally broadcasted by  $\mathcal{S}$  as the initial model, where  $\theta_g^t$  incorporates knowledge from the previous  $t-1$  tasks contributed by various clients. After  $e$  rounds of local training on  $\mathcal{T}_t^c$ ,  $\mathcal{C}_c$  obtains a new continual learning model  $\theta_{l,c}^t$  which performs well on  $K^c$ , and the corresponding average gradient update  $G^c$ .

(3) In each communication round  $r$ , the participating client  $\mathcal{C}_c$  upload its respective  $G^c$ .  $\mathcal{S}$  employs a group personalization strategy to aggregate them, which is then used to update  $A$  global models  $\theta_{g,A}^t$ . For the  $a$ -th global model  $\theta_{g,a}^t$ , it encapsulates the generalized knowledge of the associated client group's learning tasks and serves as the initial model for the next training round.

### 3.2. Memory Rehearsal in FCL

Benefiting from the efficient utilization of past experience, memory rehearsal methods [2, 10, 27] are widely used in CL as simple and effective forgetting suppression techniques. These methods involve storing partial data or knowledge from previous tasks to enable retraining during the learning of new tasks, thereby mitigating forgetting of prior tasks. Now, we extend it from CL to FCL.

**Memory Design.** Each client utilizes its limited local memory to construct a small cache space, denoted as  $M_c$ . Upon the completion of the training process for the  $t$ -th task of  $\mathcal{C}_c$ , rehearsal samples  $M_t^c$  are sampled at a fixed proportion

$\mu$  from the training set  $D_t^c$  of that task. For the initial task,  $M_0^c = \emptyset$ , and for the final task, no additional sampling operations are performed. Thus,  $M_c$  can be represented as

$$M_c = \cup_{t=1}^T M_t^c, \quad (2)$$

where the size of  $M_t^c$  is significantly smaller than  $D_t^c$ .

**Centroid-Based Representative Sample Selection.** Clearly, the quality of samples drawn from the old task dataset directly affects the quality of the memory cache. Therefore, we propose a Centroid-Based Representative Sample Selection (CBRSS) method to effectively acquire high-quality rehearsal samples. In particular, for client  $\mathcal{C}_c$ , given the distance threshold  $v$  for centroids and the input data  $x$ , the constraint condition for the minimum preset distance is defined as:

$$\min_i \|m_i^c - f_{\theta,c}(x)\| > v, \quad (3)$$

where  $m_i^c$  is the  $i$ -th centroid, and  $f_{\theta,c}(x)$  is the predictor of  $\mathcal{C}_c$  for the current task, used to map  $x$  to the feature space. If the condition in Eq. (3) is satisfied, it indicates that this data point is independent and can be considered as a new centroid  $m_{new,c} = f_{\theta,c}(x)$ . Otherwise, it means that this data point belongs to the existing centroid with the smallest distance to it, and then the target centroid is updated by

$$m_{i^*} = \frac{N_{i^*}^k \times m_{i^*} + f_{\theta,c}(x)}{N_{i^*}^k + 1}, \quad (4)$$

where  $i^* = \arg \min_i \|f_{\theta,c}(x) - m_i^c\|$ , and  $N_{i^*}^k$  represents the total number of samples in the  $k$ -th category captured by the centroid closest to  $x$  in the current task.

Overall, each client stores the representative data of each category for the trained task after completing the training. A similar approach is employed by FedMes [42], which utilizes a random sampling method to construct the memory cache. However, due to the small sample size, this approach inevitably results in significant sampling bias. The resulting  $M_t^c$  lacks representativeness and fails to reflect the overall distribution of  $D_t^c$ . This leads to a drift in the representation of old task data in the feature space, which in turn triggers the problem of catastrophic forgetting.

### 3.3. Asymmetrically Correcting Gradient Updates

To integrate the knowledge from previous tasks with that of the current task during the learning process of the  $t$ -th task on  $\mathcal{C}_c$ , we reformulate the training problem as a MOO problem, which is formally described as follows:

$$\min_{\theta_c, \theta_c^{1:t}} \ell(f_{\theta_c}, f_{\theta_c^t}, D_t^c) \cup \ell(f_{\theta_c}, f_{\theta_c^k}, M_k^c) \mid \forall k < t. \quad (5)$$

Here,  $\ell(\cdot)$  denotes the loss function, and  $\theta_c$  and  $\theta_c^t$  represent the backbone model for shared knowledge across multiple tasks and the task-specific classifier for  $\mathcal{C}_c$ , respectively.



Since the parameters of the new and old task classifiers are independent and can be updated directly using their respective gradients  $\theta_c^k \leftarrow \theta_c^k - \alpha \nabla \ell(f_{\theta_c}, f_{\theta_c^k})$  with step size  $\alpha$ , our focus is on updating the shared backbone model, which is updated using the gradients  $g_k^c = \nabla \ell(f_{\theta_c}, f_{\theta_c^k})$  of all  $t$  tasks. Let  $\beta$  be the step size for optimization. The problem of the backbone model update can be formulated as follows:

$$\theta_c \leftarrow \theta_c - \beta d_c^*, \quad d_c^* = \arg \min_{d_c} D(g_k^c, d_c) \mid \forall k \leq t, \quad (6)$$

where  $d_c^*$  represents the optimal gradient, which approximates the gradient of the new task while avoiding conflicts with the gradient of the old task. In Eq. (6), we define it as a min-distance multi-objective problem. Therefore, the critical issue lies in determining the method used to compute the differences between gradients, ensuring that the solution for  $d_c^*$  is as close as possible to all gradients.

Symmetric distance metrics, such as Euclidean distance and cosine distance, are widely applied in deep neural networks. These metrics share a fundamental property: the forward impact (from  $x$  to  $y$ ) is equal to the backward impact (from  $y$  to  $x$ ). However, local model updates are highly dependent on the magnitude and direction of gradients, both of which are inherently asymmetric. In brief, given an old task gradient  $g_{old}$ , trained on  $M_c^k$ , and a new task gradient  $g_{new}$ , trained on  $D_t^c$ , their mutual influences on each other's tasks are significantly different. In previous studies [2, 42, 49], the condition  $\langle g_{old}, g_{new} \rangle \geq 0$  has been employed to signify the absence of conflicts between two tasks. However, given that the old task has been integrated earlier,  $\|g_{old}\|$  is typically smaller while  $\|g_{new}\|$  tends to be larger. In cases where  $\|g_{old}\| \ll \|g_{new}\|$ , even if  $\langle g_{old}, g_{new} \rangle \geq 0$  holds true, the updates pertaining to the new task may have a substantial impact on the performance of the old task.

**Definition 1** (Asymmetric Gradient Distance). *Given two gradients  $g_A$  and  $g_B$ , the asymmetric gradient distance is defined as*

$$D(g_A, g_B) = \frac{\|g_A - g_B\|}{\|g_B\| + \|g_A - g_B\|}.$$

In this paper, an asymmetric distance calculation method is used to explore the measure of difference between the new task and the old task. Definition 1 considers both the gradient direction and magnitude. Notably,  $D(g_A, g_B) = 0$  when  $g_A = g_B = 0$ .

However, solving Eq. (6) directly for the parallel task flow on clients presents significant challenges. This difficulty arises from the fact that both the decision space of this problem and the model parameters are identically high-dimensional. Consequently, following [28, 35], we employ linear scalarization to determine a weight variable  $w^*$ . Specifically, when client  $C_c$  is training on the  $t$ -th task, the

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#### Algorithm 1 FedAGC

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**Input:**  $N$  clients with their private parallel task sequence  $\mathcal{T}^{1:N}$  and local datasets  $\{D_t^{1:N}\}_{t=1}^T$ , global model parameter  $\theta_g$ , communication rounds  $R$ , memory space  $M_{1:c} \leftarrow \{\}$   
**Output:** Multiple global parameters  $\theta_g^{i:n}$ , local parameter  $\theta_{l,1:N}$  and classification head  $\theta_{l,1:N}^k$

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1: Initialization
2: while task number  $t \leq T$  do
3:   for each round  $r = 1, 2, \dots, R$  do
4:     Server broadcasts  $\theta_g^i$  to each group of clients;
5:     for each client  $c = 1$  to  $N$  in parallel do
6:       Minimizing local CL problems by Eq. (1);
7:        $g^c = \nabla \ell(f_{\theta_c}, f_{\theta_c^t}, D_t^c)$ ;
8:        $g_k^c = \nabla \ell(f_{\theta_c}, f_{\theta_c^k}, M_k^c)$  for all  $k < t$ ;
9:        $d_c^* \leftarrow \text{ACGU}(g^c, g_1^c, \dots, g_{t-1}^c)$ ;
10:       $\theta_{l,c}^t \leftarrow \theta_{l,c}^t - \alpha d_c^*$ ;
11:    end for
12:    Send the  $G^c = -\Delta \theta_{l,c}^t / \eta$  to the central server;
13:     $\{S_1, S_2, \dots, S_n\} \leftarrow \text{GAGU}(G^1, G^2, \dots, G^N)$ ;
14:     $\theta_g^i \leftarrow \theta_g^i - \frac{1}{|S_i|} \sum_{j \in S_i} G^j$  for  $i \in \{1, 2, \dots, n\}$ ;
15:  end for
16:  Each client samples  $M_t^c$  from  $D_t^c$  by Eqs. (3) and (4);
17: end while

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gradient update  $g_t^c$  of the new task and  $g_{1:t-1}^c$  from the old tasks and computed through

$$w^* = \arg \min_w \left\{ \frac{1}{t} \sum_i^t D \left( \sum_{j \neq i}^t w_j g_j^c, g_i^c \right) \right\}, \quad (7)$$

where  $\sum_{j=1}^t w_j = 1$ ,  $\forall w_j \geq 0$ , and  $D(\sum_{j \neq i}^t w_j g_j^c, g_i^c)$  represents the divergence between task  $i$  and the combination of all other tasks. By minimizing this divergence, we achieve the suppression of the influence from the combination on task  $i$ . Note that this optimization process does not modify individual gradient updates.

To prevent training failure on new tasks, we introduce the Multiple Gradient Descent Algorithm (MGDA) [4] to minimize the gradient norm, thereby achieving a balanced solution. It is defined as

$$w_b^* = \arg \min_w \left\| \sum_i^t w_i g_i^c \right\|^2. \quad (8)$$

When the objective function value of MGDA is zero, the obtained solution satisfies the Karush-Kuhn-Tucker conditions. Otherwise, the solution provides a Pareto descent direction that does not harm any task.

In summary, our proposed Asymmetric Correction Gradient Updating (ACGU) algorithm determines  $d_c^*$  by

$$d_c^* = \sum_i W_i g_i^c, \quad \text{where } W = \gamma w^* + (1 - \gamma) w_b^*. \quad (9)$$

| Algorithm      | Attributes |     |     | Task ID under Non-IID Setting (Across 8 Different Datasets) |                     |                     |                     |                     | $A_{avg}$ (%)       |
|----------------|------------|-----|-----|-------------------------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                | FCL        | Per | GCB | $A_1$ (%)                                                   | $A_2$ (%)           | $A_3$ (%)           | $A_4$ (%)           | $A_5$ (%)           |                     |
| FedAvg         | ×          | ×   | ×   | 54.73 ± 1.48                                                | 56.06 ± 3.16        | 63.99 ± 3.43        | 57.90 ± 2.96        | 78.49 ± 2.18        | 61.83 ± 0.46        |
| FedProx        | ×          | ×   | ×   | 58.25 ± 1.60                                                | 56.65 ± 0.50        | 66.15 ± 2.33        | 61.72 ± 2.21        | 82.24 ± 1.02        | 65.00 ± 0.62        |
| Ditto          | ×          | ✓   | ×   | 58.92 ± 1.40                                                | 59.13 ± 1.63        | 70.58 ± 3.07        | 63.00 ± 2.79        | 83.97 ± 0.95        | 67.12 ± 0.85        |
| FedCurv        | ✓          | ×   | ✓   | 60.18 ± 3.43                                                | 54.88 ± 1.99        | 68.05 ± 3.78        | 59.25 ± 2.37        | 82.19 ± 0.58        | 64.91 ± 0.56        |
| FedAGEM        | ✓          | ×   | ✓   | 69.62 ± 0.85                                                | 66.91 ± 1.09        | 75.06 ± 1.67        | 66.73 ± 2.47        | 81.73 ± 1.16        | 72.01 ± 0.81        |
| FedER          | ✓          | ×   | ×   | 73.34 ± 0.45                                                | 68.77 ± 1.89        | 80.18 ± 1.39        | 70.19 ± 1.30        | 79.43 ± 1.32        | 74.38 ± 0.27        |
| FedMes         | ✓          | ✓   | ✓   | 70.84 ± 0.79                                                | 67.22 ± 1.05        | 77.25 ± 0.89        | 67.18 ± 1.52        | 85.21 ± 0.36        | 73.54 ± 0.55        |
| Ours-w/o-ACGU  | ×          | ✓   | ×   | 58.62 ± 0.36                                                | 58.06 ± 0.16        | 65.62 ± 0.19        | 63.29 ± 1.86        | 82.46 ± 1.68        | 65.61 ± 0.80        |
| Ours-w/o-GAGU  | ✓          | ×   | ✓   | 76.19 ± 0.52                                                | 71.59 ± 1.12        | 82.24 ± 1.42        | 72.89 ± 2.04        | 81.82 ± 2.36        | 76.95 ± 1.38        |
| Ours-w/o-CBRSS | ✓          | ✓   | ✓   | 75.40 ± 0.90                                                | 72.39 ± 0.55        | 82.88 ± 0.92        | 75.42 ± 0.54        | 86.68 ± 0.62        | 78.56 ± 0.57        |
| FedAGC (Ours)  | ✓          | ✓   | ✓   | <b>76.21 ± 0.36</b>                                         | <b>72.56 ± 0.48</b> | <b>84.90 ± 0.85</b> | <b>77.60 ± 0.57</b> | <b>87.28 ± 0.18</b> | <b>79.71 ± 0.23</b> |

Table 1. Performance (avg ± std) of local models for each task on 10 clients under Non-IID setting. "w/o-Module" indicates performance without the corresponding module. It also includes Per (Personalized) and GCB (Gradient Correction-Based). For the proposed FedAGC, we used  $\lambda = 0.01$  and  $\gamma = 0.5$  in Eqs. (1) and (9).

### 3.4. Group-Level Personalization

To integrate heterogeneous knowledge from different clients, some studies [39, 45] adopt personalization strategies at the client level, where each client continuously adjusts the received global model to better fit its local tasks. However, this approach struggles to meet the needs of all clients in large-scale parallel task scenarios. In contrast, we consider a group-level personalization strategy, enabling the broadcast of global models tailored to meet the specific requirements of each client.

In this work, we propose Group Aggregation Gradient Updating (GAGU) for client-uploaded gradients. Specifically, during the  $r$ -th communication round,  $N$  clients upload their respective averaged gradient updates  $\{G^1, G^2, \dots, G^N\}$  obtained after  $e$  local training iterations, where  $G^c = -\Delta\theta_{l,c}^t/\alpha$ . We compute a similarity matrix  $S$  to measure the similarity between pairs of averaged gradient updates. The elements of  $S$  are given by

$$s_{i,j} = \frac{G^i G^j}{\|G^i\| \|G^j\|}. \quad (10)$$

We construct a graph  $P = \{G, E\}$  to represent  $A$ , where  $G$  denotes the set of nodes in the graph, and each node corresponds to the  $G^c$ . The set of edges  $E = \{s_{i,j}\} \forall i \leq j \leq N$  represents the similarity between  $G^i$  and  $G^j$ . The objective of GAGU is to reorder  $P$  such that similar average gradient updates are grouped together, while dissimilar ones are isolated. To achieve this, we first define the similarity between any two groups  $S_A$  and  $S_B$  ( $S_A \cap S_B = \emptyset, \forall A \neq B$ ):

$$h(S_A, S_B) = \sum_{i \in S_A, j \in S_B} s_{i,j}. \quad (11)$$

GAGU aims to maximize intra-group similarity while minimizing inter-group similarity, resulting in an optimal graph partition  $a^* = \{S_1, S_2, \dots, S_n\}$ , where  $n$  represents the

number of groups in  $a^*$ . Inspired by spectrum-clustering [13] and Ncut [36], We reformulate this optimization problem into the following form:

$$\min_{a \in A} \sum_{S, S' \in a} h(S, S') - h(S', S'), \quad (12)$$

where  $A$  is all possible grouping results. Additionally, we introduce the modularity  $Q$  to evaluate the quality of the current partition during the process of determining the optimal number of groups and partitioning results. This metric measures the distribution of connectivity within and between groups.

$$Q = \frac{1}{2W} \sum_{i,j} \left[ s_{i,j} - \frac{d_i d_j}{2W} \right] \delta(S_i, S_j), \quad (13)$$

where  $W = \sum_{i,j} s_{i,j}$  represents the total edge weight in the current graph,  $d_i = \sum_j s_{i,j}$  denotes the degree of node  $i$ , and  $\delta(S_i, S_j)$  is the Kronecker delta function.

## 4. Experiments

### 4.1. Experimental Setup

**Benchmarks.** We validate FedAGC in large-scale multi-task parallel FCL scenarios. The experimental setup follows the Task-Incremental Learning paradigm and primarily includes two different benchmarks: (1) **IID Setting (Single CIFAR-100)**: The 100 classes of CIFAR-100 [16] are divided into 20 tasks, each containing 5 classes. These tasks are randomly assigned to clients, ensuring that each client does not train on the same task multiple times. (2) **Non-IID Setting (Across 8 Different Datasets)**: We partition and distribute eight widely used datasets in the FCL domain, such as CIFAR-10/100 [16], MNIST [18], SVHN [30], FashionMNIST [41], Not-MNIST [1], FaceScrub [32], and TrafficSigns [31], in the same manner as (1). In both benchmarks, each client is assigned a unique task sequence.

| Algorithm      | Attributes |     |     | Task ID under IID Setting (Single CIFAR-100) |                     |                     |                     |                     | $A_{avg}$ (%)       |
|----------------|------------|-----|-----|----------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                | FCL        | Per | GCB | $A_1$ (%)                                    | $A_2$ (%)           | $A_3$ (%)           | $A_4$ (%)           | $A_5$ (%)           |                     |
| FedAvg         | ×          | ×   | ×   | 54.32 ± 0.94                                 | 60.53 ± 2.08        | 61.89 ± 1.56        | 65.63 ± 1.60        | 70.77 ± 0.84        | 62.63 ± 0.55        |
| FedProx        | ×          | ×   | ×   | 53.96 ± 0.98                                 | 61.53 ± 2.23        | 62.90 ± 1.94        | 65.70 ± 1.13        | 70.78 ± 0.91        | 62.97 ± 0.93        |
| Ditto          | ×          | ✓   | ×   | 55.48 ± 0.92                                 | 62.81 ± 1.96        | 63.12 ± 1.58        | 66.59 ± 1.86        | 71.01 ± 0.93        | 63.67 ± 0.72        |
| FedCurv        | ✓          | ×   | ✓   | 54.36 ± 0.95                                 | 60.77 ± 0.86        | 63.43 ± 0.91        | 66.46 ± 0.97        | 71.52 ± 0.75        | 63.31 ± 0.16        |
| FedAGEM        | ✓          | ×   | ✓   | 60.95 ± 0.90                                 | 67.15 ± 1.23        | 66.39 ± 0.69        | 67.81 ± 0.90        | 71.28 ± 0.66        | 66.72 ± 0.23        |
| FedER          | ✓          | ×   | ×   | 63.05 ± 0.76                                 | 67.83 ± 0.91        | 66.11 ± 0.32        | 65.39 ± 0.44        | 68.84 ± 1.01        | 66.44 ± 0.42        |
| FedMes         | ✓          | ✓   | ✓   | 61.18 ± 0.34                                 | 67.68 ± 0.11        | 67.11 ± 0.49        | 67.42 ± 0.15        | 72.12 ± 0.90        | 67.13 ± 0.16        |
| Ours-w/o-ACGU  | ×          | ✓   | ×   | 55.45 ± 0.27                                 | 61.95 ± 0.87        | 63.32 ± 1.72        | 65.51 ± 0.73        | 71.44 ± 0.60        | 63.53 ± 0.31        |
| Ours-w/o-GAGU  | ✓          | ×   | ✓   | 63.92 ± 0.55                                 | 66.73 ± 0.93        | 67.22 ± 0.61        | 66.48 ± 0.65        | 68.53 ± 0.24        | 66.58 ± 0.41        |
| Ours-w/o-CBRSS | ✓          | ✓   | ✓   | 66.14 ± 0.37                                 | 69.05 ± 0.80        | 69.69 ± 0.49        | 69.56 ± 2.20        | <b>74.12 ± 0.09</b> | 69.75 ± 0.78        |
| FedAGC (Ours)  | ✓          | ✓   | ✓   | <b>66.38 ± 0.48</b>                          | <b>69.48 ± 0.78</b> | <b>70.38 ± 1.32</b> | <b>70.58 ± 1.85</b> | 73.98 ± 0.41        | <b>70.16 ± 0.77</b> |

Table 2. Performance (avg ± std) of local models for each task on 10 clients under IID setting.

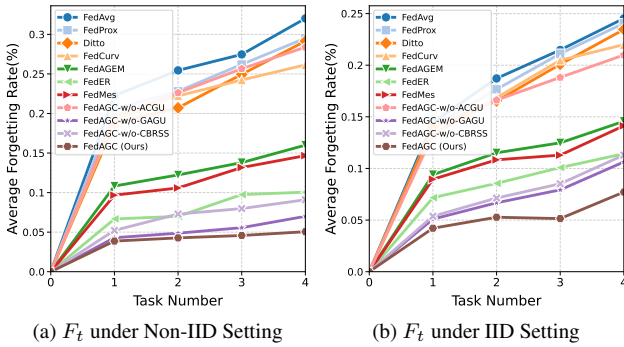


Figure 3. Comparison of average forgetting rates.

**Baselines.** We compare FedAGC with state-of-the-art (SOTA) methods, including the fundamental FedAvg [29] and its proximal variant FedProx [19], along with the personalized FL method Ditto [20]. To verify the superiority of FedAGC, we also compare it with gradient correction-based FCL methods, including FedCurv [38], FedAGEM [2] and the latest FedMes [42]. Additionally, we consider the rehearsal-based method FedER [3].

**Metrics.** We employ two critical metrics to evaluate algorithm performance: Average Accuracy ( $A_{avg}$ ) and Forgetting Rate ( $F_t$ ). Let  $a_{i,t}^c$  denote the test accuracy of the  $i$ -th task learned by the  $c$ -th client after sequential learning of  $t$  subsequent tasks, where  $t + i \leq T$ . In this formulation, the initial accuracy of task  $i$  (immediately after its own training) and the final accuracy after training on all tasks are defined as  $a_{i,0}^c$  and  $a_{i,T-i}^c$ , respectively. The forgetting rate for the  $i$ -th task after learning  $t$  subsequent tasks is calculated as:  $f_{i,t}^c = \frac{a_{i,0}^c - a_{i,t}^c}{a_{i,0}^c}$ . To aggregate results, we compute client-wise averages:  $A_i = \frac{1}{N} \sum_{c=1}^N a_{i,T-i}^c$  and  $f_{i,t} = \frac{1}{N} \sum_{c=1}^N f_{i,t}^c$ . Then,  $A_{avg}$  and  $F_t$  are defined as:

$$A_{avg} = \frac{1}{T} \sum_{i=1}^T A_i, \quad F_t = \frac{1}{T-t} \sum_{i=1}^{T-t} f_{i,t}. \quad (14)$$

**Implementation details.** Following the experimental setup

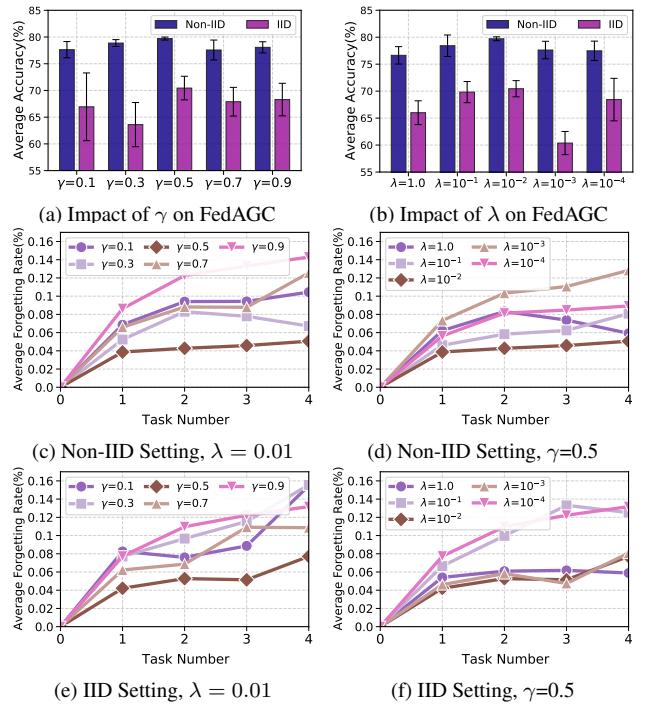


Figure 4. Impact of core hyperparameters on FedAGC.

of [47], we adopted a variant of the LeNet architecture as the backbone. For any newly introduced task, we initialized a new classifier. The Adam optimizer was used for all datasets, with an initial learning rate of  $10^{-3}$  and a weight decay of  $10^{-4}$ . The experimental setup consisted of 10 clients, each learning 5 sequential and non-overlapping tasks. The batch size was set to 100 per step, with  $\mu$  and  $v$  set to 0.1 and 8, respectively. To ensure convergence, all experiments were conducted for 100 federated communication rounds. Each experiment was repeated five times, and the reported results were averaged accordingly.

## 4.2. Main Results

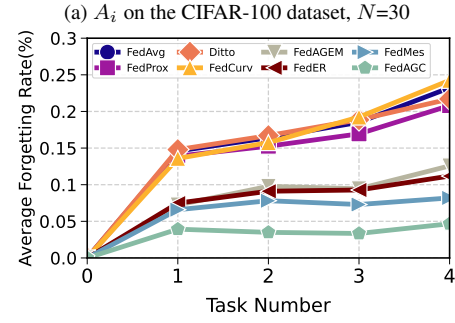
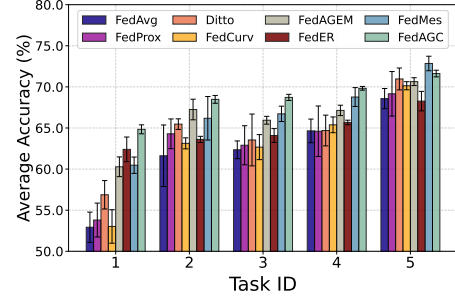
Tabs. 1 and 2 record the average accuracy of 10 clients on each task under two data condition setups, along with the

final results considering all tasks. In Fig. 3, we compare the degree of Spatial-TemporalCF between FedAGC and various methods. It can be observed that FedProx and Ditto mitigate the impact of statistical heterogeneity using penalty terms, showing significant performance improvements over FedAvg in Non-IID settings. However, since they do not employ any CL strategies, their accuracy is lower compared to FCL methods such as FedAGEM, FedER, and FedMes, with higher forgetting levels. Notably, FedCurv’s experimental results are suboptimal as it leverages curvature information from other clients to guide updates for the current client, which is prone to knowledge conflicts in cases of non-overlapping client data distributions. These FCL methods based on symmetric gradient correction focus solely on TemporalCF without considering the interference of knowledge across clients, leading to SpatialCF. In contrast, the proposed FedAGC outperforms other methods in both task-specific accuracy and overall performance across both settings, with a lower degree of forgetting. This demonstrates that memory buffers constructed through importance sampling, assisted by asymmetric gradient correction and group-level personalization, can significantly improve the accuracy of PFCL models.

### 4.3. Performance Analysis of FedAGC

**Ablation Studies.** We conducted three different ablation studies under the same experimental setup, with the results recorded in Tabs. 1 and 2, as well as Fig. 3. We progressively removed components from FedAGC, including: (1) Omitting the ACGU module (FedAGC-w/o-ACGU), which means that only the gradients from local training are used to guide updates. Clearly, in this case, the CBRSS module also loses its significance. The performance degradation is significant, with a drop of 17.7% and 9.5%, as no strategy for mitigating TemporalCF is applied. (2) Removing the GAGU module (FedAGC-w/o-GAGU) and switching to the FedAvg average aggregation method. In the IID setting, FedAGC-w/o-GAGU’s overall performance is overtaken by FedMes by 0.86%. (3) Disabling the CBRSS module (FedAGC-w/o-CBRSS) and replacing it with a random sampling method. In this case, performance decreases by 1.44% and 0.58% in the Non-IID and IID settings, respectively. However, since the core component for mitigating forgetting is retained, its accuracy on individual tasks and overall performance remains superior to other baselines. These results verify that our three proposed modules of FedAGC play a crucial role in alleviating Spatial-TemporalCF and achieving personalization, and can effectively be combined to enhance performance.

**Impact of core hyperparameters.** We evaluate FedAGC with different values of  $\gamma$  and  $\lambda$ . As shown in Fig. 4, setting  $\lambda$  too large or too small causes FedAGC to overly rely on either special or balanced solution, impairing its ability to



(a)  $A_i$  on the CIFAR-100 dataset,  $N=30$   
(b)  $F_t$  on the CIFAR-100 dataset,  $N=30$

Figure 5. Performance on large-scale clients.

mitigate forgetting. Similarly, an excessively large  $\gamma$  biases the model toward the inversion of previous tasks, hindering the learning of current task, while a small  $\gamma$  has the opposite effect. Empirically, we find that the best performance of FedAGC is achieved when  $\gamma = 0.5$  and  $\lambda = 0.01$ .

**Performance on large-scale clients.** To explore the scalability of FedAGC to larger client scenarios, we tripled the number of clients from the previous setting of 10 and continued comparing it with various SOTA methods. The results recorded in Fig. 5 indicate that FedAGC outperforms other baselines in accuracy on previous tasks and demonstrates greater stability in mitigating forgetting. This result highlights the superiority of FedAGC.

## 5. Conclusion

To tackle the challenge of STCF in PFCL, this paper proposed FedAGC as a valid solution. After local training, FedAGC selects representative samples and stores them in a buffer for rehearsal to alleviate forgetting. To refine this process, we introduced an asymmetric gradient metric to quantify discrepancies between past and current tasks. Asymmetric gradient correction is then employed to adjust the gradient updates, jointly considering direction and magnitude for optimal adaptation. Moreover, FedAGC adopts group-level personalization by clustering clients based on the similarity of their uploaded averaged gradient updates, which enhances the fusion of valuable knowledge while isolating ineffective knowledge. It was extensively demonstrated in this paper that the effective integration of the proposed modules can enhance the model’s resistance to STCF, enabling FedAGC to outperform SOTA approaches.



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